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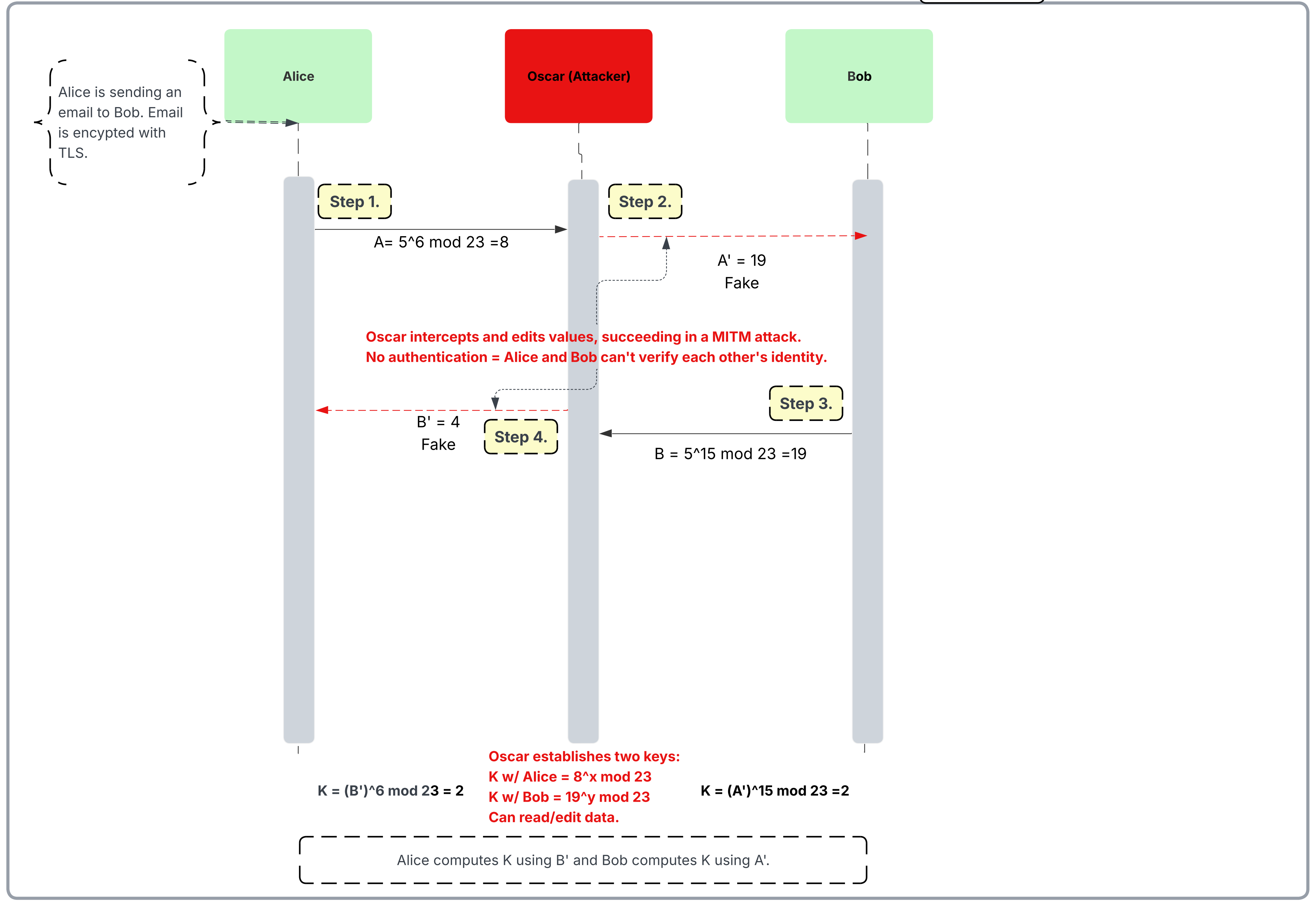
CS 3110
MITM Diagram
4/17/26

Diagram key

- ● Normal Communication (black arrows)
- 🟩 Recipients
- ● Data
- 🟥 Attacker / edited messages (dotted red arrows)

Diffie-Hellman Key Exchange MITM diagram

Public Values:
p=23
g=5



Alice is sending an email to Bob. Email is encypted with TLS.

Oscar intercepts and edits values, succeeding in a MITM attack.
No authentication = Alice and Bob can't verify each other's identity.

K = (B')^6 mod 23 = 2

Oscar establishes two keys:
K w/ Alice = 8^x mod 23
K w/ Bob = 19^y mod 23
Can read/edit data.

K = (A')^15 mod 23 =2

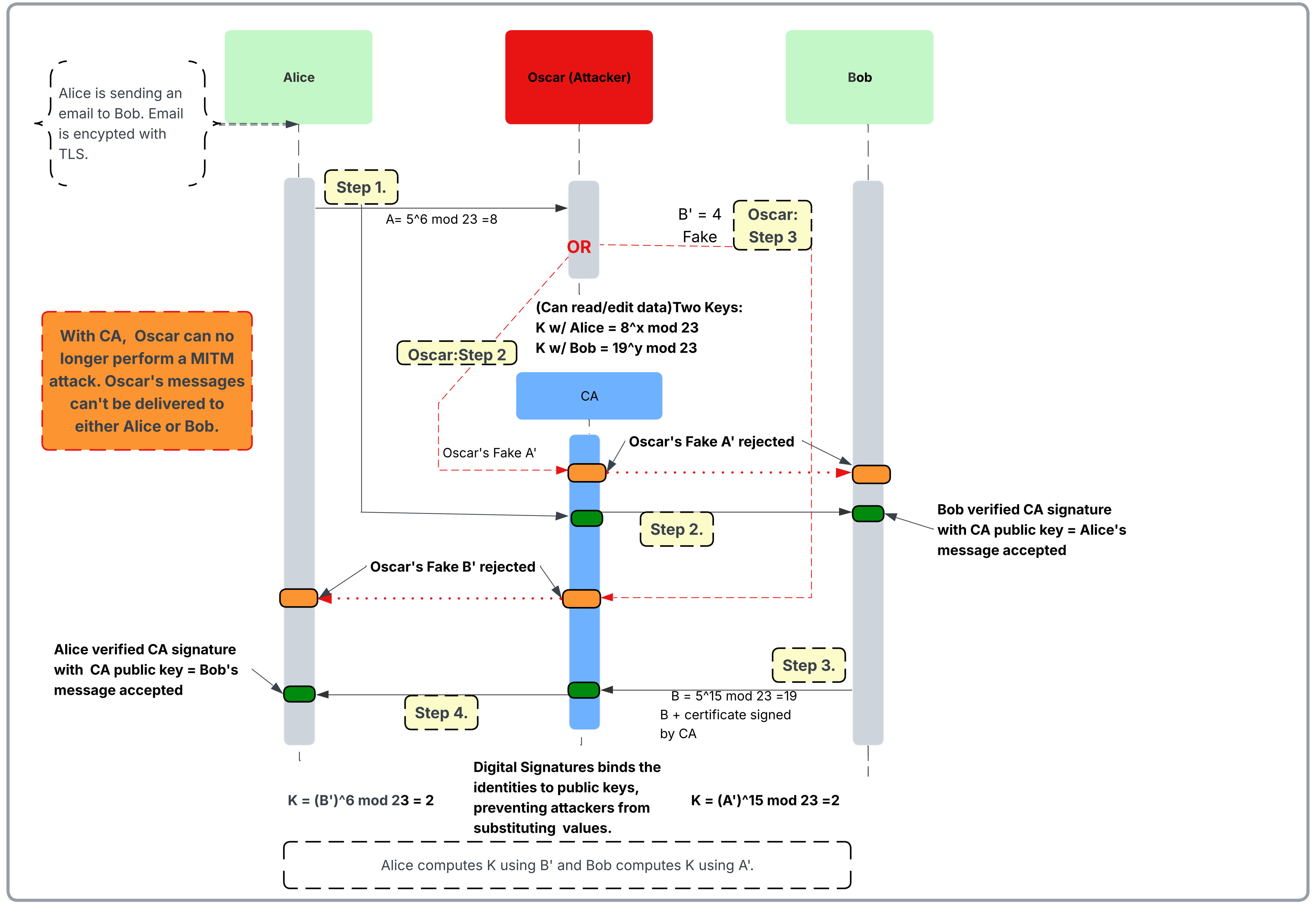
Alice computes K using B' and Bob computes K using A'.

Diagram key

- ● Normal Communication (black arrows)
- 🟩 Recipients
- ● Data
- 🟥 Attacker / Edited Messages
- 🟦 Certificate/ Signed Message
- 🟢 CA Signature Verification
- 🟠 Rejected Message

Diffie-Hellman Key Exchange MITM w/ CA diagram

Public Values:
p=23
g=5



Alice is sending an email to Bob. Email is encypted with TLS.

With CA, Oscar can no longer perform a MITM attack. Oscar's messages can't be delivered to either Alice or Bob.

(Can read/edit data)Two Keys:
K w/ Alice = 8^x mod 23
K w/ Bob = 19^y mod 23

Bob verified CA signature with CA public key = Alice's message accepted

Alice verified CA signature with CA public key = Bob's message accepted

B = 5^15 mod 23 =19
B + certificate signed by CA

K = (B')^6 mod 23 = 2

Digital Signatures binds the identities to public keys, preventing attackers from substituting values.

K = (A')^15 mod 23 =2

Alice computes K using B' and Bob computes K using A'.